Available online at www.sciencedirect.com
SciVerse ScienceDirect
Current Opinion in Neurobiology

Neural specializations for small target detection in insects

Karin Nordström

Despite being equipped with low-resolution eyes and tiny brains, many insects show exquisite abilities to detect and pursue targets even in highly textured surrounds. Target tracking behavior is subserved by neurons that are sharply tuned to the motion of small high-contrast targets. These neurons respond robustly to target motion, even against self-generated optic flow. A recent model, supported by neurophysiology, generates target selectivity by being sharply tuned to the unique spatiotemporal profile associated with target motion. Target neurons are likely connected in a complex network where some provide more direct output to behavior, whereas others serve an inter-regulatory role. These interactions may regulate attention and aid in the robust detection of targets in clutter observed in behavior.

Address
Department of Neuroscience, Uppsala University, Box 593, 751 24 Uppsala, Sweden

Corresponding author: Nordström, Karin (karin.nordstrom@neuro.uu.se)

Current Opinion in Neurobiology 2012, **22**:272–278

This review comes from a themed issue on Neuroethology
Edited by Michael Dickinson and Cynthia Moss

Available online 12th January 2012

0959-4388/$ – see front matter
© 2012 Elsevier Ltd. All rights reserved.

DOI 10.1016/j.conb.2011.12.013

Introduction

As an animal moves through the world its own motion generates optic flow across the retina. Such optic flow cues can be used to maintain a straight heading or to avoid obstacles. Besides optic flow generated by ego-motion, many animals can also discriminate the motion of objects that move independently of the remaining visual surround. The detection of widefield optic flow and of local target motion, respectively, constitute two fundamental key components of motion vision that biological visual systems need to discriminate and process.

Successful visualization of small targets that move rapidly across the visual field would intuitively require high spatial resolution, as provided in the single lens eye of vertebrates. As a consequence of the high resolution, some birds of prey, such as kestrels, can strike targets from an astonishing distance of 275 m [1]. However, not only vertebrates, but also many insects, such as dragonflies and hoverflies, perform elaborate and elegant high-speed pursuits of small targets, representing prey, predators or conspecifics [(e.g. 2–4)]. This is impressive considering that insects have tiny brains, and that they carry compound eyes with inherently poor spatial resolution [5]. Many insects use behavioral adaptations to improve target visualization, such as viewing the target against the clear sky [6] to increase relative contrast, or adopting a hovering [7] or a perching stance [3], thereby rendering the background stationary. Nevertheless, during actual pursuit the target will inevitably be displayed against the pursuer's self-generated optic flow. For successful capture the pursuer thus has to visualize the target's motion against a moving background.

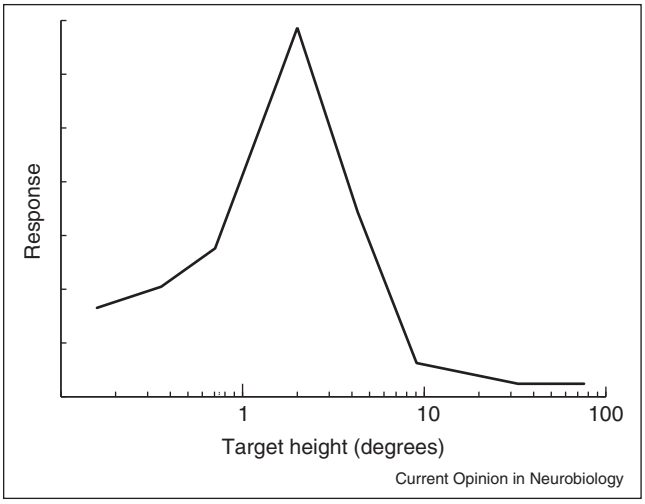
Both insects and vertebrates have higher-order visual neurons tuned to the motion of small targets [4,8–11]. In this review I will discuss the physiology of such target tuned neurons in hoverflies and dragonflies, in light of a recent model for small target detection, and how this model compares with a common model for optic flow detection. I will also discuss how target neurons may regulate the activity of each other in a complex network.

Target neurons and their inputs

Behavioral target detection is likely subserved by small target motion detectors (STMDs) located in the third optic ganglion, the lobula. STMDs are identified by their sharp size tuning: they give peak responses to targets subtending 1–3° of the visual field, with no response to larger bars (typically > 10°, Figure 1) or to widefield grating stimuli [12••]. Intriguingly, despite the vast differences between the vertebrate and insect optical input, STMDs bear striking similarity to the classic hypercomplex cells [11] of the vertebrate visual cortex [(for review, see 13)].

How is target selectivity generated at the neuronal level? Classic models rely on subtracting the output from neurons sensitive to the optic flow generated by the viewer's ego-motion, making target motion the only remaining signal [14]. Such negative feedback is not only intuitive and computationally concise, it has also been shown to underlie figure selectivity in figure detection (FD) neurons in the blowfly [15,16]. However, such an inhibitory mechanism leads to neurons that fail to respond if a target is visualized against a background that is itself moving [17]. During actual behavioral pursuit targets will often be displayed against self-generated optic flow. More suitable for such behavior, several STMDs continue to respond robustly to moving targets, whether these are displayed against a bright background or a high-contrast textured background, and whether or not there is a

Figure 1



Sharp size tuning of STMDs. The graph shows the response of an STMD to targets of varying height.

relative velocity difference between the target and the background [12••,18•]. This is a striking finding, considering the importance relative motion models have been given in the literature [(e.g. 14,17,19)].

A recent model [20••] explains how targets can be visualized against moving clutter, even without relative velocity differences (Figure 2a). The model relies on the signature of a dark target moving across a single point in space. Such motion has a unique spatiotemporal profile where the spatially limited target's leading edge (dark contrast change) is followed a short time later by its trailing edge (bright contrast change). The core of Wiederman's model [20••] thus depends on half-wave rectification that splits the signal from the same point in space into separate ON and OFF contrast channels, corresponding to the leading and trailing edges of the target, which are then temporally correlated (Figure 2a). The addition of fast adaptation and strong spatial antagonism inside the motion detector itself leads to sensitivity to the spatiotemporal dynamics associated with target motion. The resulting non-directional elementary STMDs can be spatially pooled in STMDs with larger receptive fields [20••].

Early anatomical studies of the fly optic lobes showed that the column underlying each facet is represented by up to 100 unique interneurons, leading to the suggestion that visual input is processed in many parallel streams [21–23,24•]. In support of this notion, local motion is computed in a fundamentally different way in the elementary SMTD ([20••], Figure 2a), compared with the classic correlation type elementary motion detector (EMD [25]). In the EMD, directional motion sensitivity is generated by temporally correlating the luminance input from two neighboring points in space. EMD input underlies widefield optic flow processing in most biological systems [26]. Whereas electrophysiology on neurons sensitive to optic flow, and behavioral responses to similar stimuli, provide strong support in favor of an underlying EMD input, recent technical advances have allowed more direct investigation of the EMD itself, in the genetic model *Drosophila* [27]. Even if the precise layout is still under intense debate [28–30], recent work suggests that the EMD input may also be halfwave rectified [31], and split into separate ON and OFF channels [32•] (the 2-quadrant model [29,33] shown in Figure 2b).

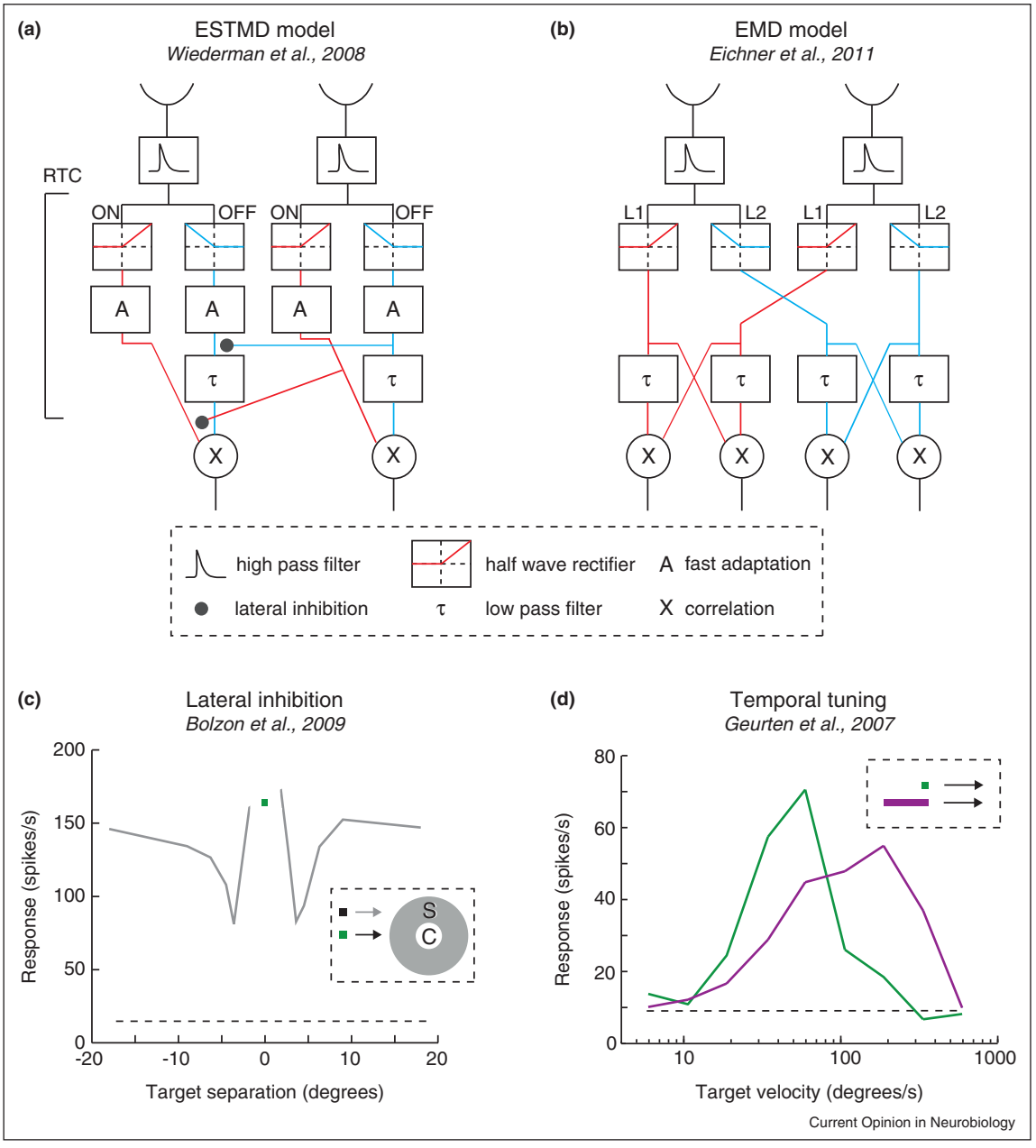
Target detection using spatiotemporal dynamics

Several predictions of the elementary STMD model have been confirmed in electrophysiological recordings of their downstream targets, the STMDs. For example, the model relies on strong spatial inhibition from neighboring units (Figure 2a). The presence of such lateral inhibition was investigated by quantifying the response to a target moving in the presence of a distracter target at varying distances [34]. As the distracter target moved through the strong symmetrical surround, peak inhibition was generated at ca. 3° separation (Figure 2c). This matches the predictions from the model [20••], and thus accounts for the sharp size tuning of STMDs (Figure 1).

Besides the target's spatial profile, elementary STMD output relies strongly on the target's temporal profile [20••]. STMDs are velocity tuned [35,36], with peak responses to velocities matching the temporal filters of the model [20••]. Seen from a single point in space, a wide target moving fast approaches the temporal profile of a narrow target moving slower. Responses of a dragonfly STMD to targets that were either square ($0.8 \times 0.8°$) or elongated ($0.8 \times 8°$) showed that the velocity optimum had in fact shifted to higher velocities for the wider targets [35], highlighting the important role temporal mechanisms play (Figure 2d).

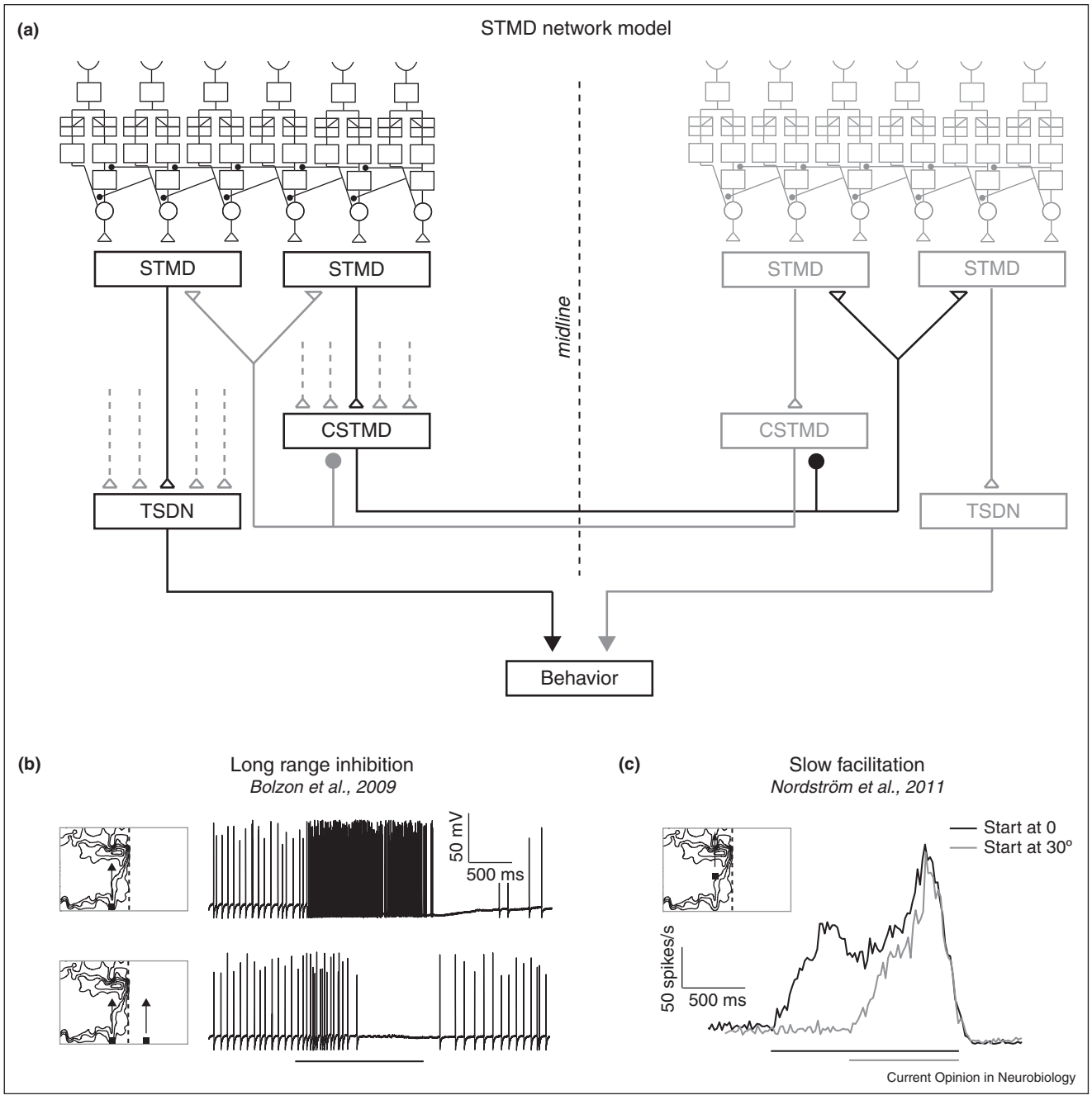
To function during actual target pursuit, STMDs need to respond to a target visualized against background motion and not to features embedded in the background texture. Accompanying this requirement, elementary STMD modeling [20••] and STMD physiology [18•] show that branches, edges and other features that may approach the correct spatiotemporal profile of an optimal target are surprisingly rare in natural scenes. Furthermore, the responses to such false-positive targets are much smaller than responses to optimal high-contrast targets inserted in conspicuous spots of a panoramic natural scene [18•]. These examples (response to natural scenes, Figure 2c,d) thus suggest that the elementary STMD model [20••] provides a good computational framework for explaining how small target selectivity is generated in STMDs.

Figure 2



Elementary STMD model and predictions. **(a)** The elementary STMD (ESTMD) model [20••] provides a framework for how small target selectivity may be generated. It is based on the physiology of rectified transient cells (RTCs) in the fly medulla, and contains the following key components: the signal from early visual processing is high-pass filtered and half-wave rectified into separate ON and OFF channels. The ON and OFF channels adapt rapidly (A) and are each subjected to spatial lateral inhibition from neighboring units (only one shown here, for simplicity) of the same contrast polarity (i.e. $ON_{surround} \rightarrow ON_{center}$, and $OFF_{surround} \rightarrow OFF_{center}$). The OFF signal is delayed via a low-pass filter (τ) before being correlated (X) with the ON signal. The original model [20••] includes several stages and elaborations, particularly at earlier stages of visual processing, not shown here. **(b)** The classic elementary motion detector (EMD) has recently received renewed attention. Shown here is the so-called 2-quadrant version of the model [29]. Two neighboring inputs are each subjected to high-pass filtering and half-wave rectification into separate ON and OFF channels (via the lamina monopolar cells L1 and L2, respectively). The signal from one ON input is delayed via a low-pass filter (τ) before being correlated (X) with the ON signal from a neighboring input. Similarly, the OFF signal from one input is delayed and correlated with the OFF signal from a neighboring input. The outputs from the two half-arms are then subtracted to generate direction-selectivity (not shown). **(c)** The presence of spatial, lateral inhibition (see pictogram, C = center, S = surround) was confirmed in electrophysiological recordings of a dragonfly STMD [34]. By displaying a primary target (green) in the presence of a distracter target (black), we showed symmetrical peak inhibition at ca. 3° separation. Negative values on the *x*-axis indicate that the distracter target moved below the primary target, whereas positive values indicate that it moved above. **(d)** The importance of the temporal profile of target motion was confirmed by displaying targets that were either 0.8° square (green), or 0.8 × 8° wide (purple), and measuring the response as a function of velocity. The data [35] show that the peak response is shifted to a higher velocity for the wider targets.

Figure 3



STMD network model. **(a)** The output of many elementary STMDs can be spatially pooled in STMDs with larger receptive fields, but the same position invariant sharp size tuning as the elementary STMDs themselves [20••]. STMDs, in turn, may synapse with target-selective descending neurons (TSDNs), or with centrifugal STMDs (CSTMDs) that cross the midbrain and provide output to the contralateral hemisphere, where they may inhibit the response of their contralateral counterpart, or affect other motion sensitive neurons. The dashed lines indicate that input may be pooled from several STMDs to a single CSTMD, or TSDN. **(b)** The presence of long-range inhibition acting between the two CSTMD1 neurons in each hemisphere was recently confirmed [34]. When a single target moves through the center of the receptive field, CSTMD1 responds with a strong burst of action potentials (top trace). If we simultaneously display a second target in a mirror symmetric position (dashed line indicates the animal's midline), the response is completely inhibited (lower trace). The bar indicates the peri-stimulus duration. **(c)** CSTMD1 has an unusually long response onset. The black trace shows the response to a target starting its trajectory at the bottom of the screen (as shown in the top pictogram in panel b). The gray trace shows the response to a target starting 30° higher up, within the receptive field (see pictogram). It takes several hundred milliseconds for the response to reach the levels of the space-aligned control [39]. The bars indicate the peri-stimulus durations of the two trajectories, respectively.

Target neurons and their connections

While we are starting to understand the computational mechanisms that underlie the exquisite target selectivity of STMDs, we still know little about how these neurons contribute to observed behavior. Dragonflies and male hoverflies have about 20 different STMDs [9,12••], which vary in terms of receptive field size and location, directionality, and ability to discriminate targets in clutter. Some STMDs output in the lateral subesophageal ganglion [12••], where they likely synapse with target-selective descending neurons (TSDNs [13,37]), which control dragonfly wing movement [4,10]. Other STMDs output in the lateral midbrain [38] where they could synapse with centrifugal STMDs (CSTMDs).

The morphology of STMDs, albeit limited, thus suggests that STMDs are connected to each other in a complex network (simplified schematic, Figure 3a), where some have a more direct effect on motor output, and others fine-tune the responses of other STMDs. Whereas direct evidence for STMD–STMD interactions remains limited, some hints to the presence of intriguing network interactions are available from the dragonfly CSTMD1, which has two output arbors in the contralateral hemisphere [35]. One of these outputs synapse with the input of its contralateral counterpart, inhibiting its response (Figure 3a). This heterolateral inhibition was shown by comparing the response to a single target traversing the CSTMD1 receptive field, which then responds with a vigorous burst of action potentials (top trace, Figure 3b). If instead two targets are displayed, with the second target moving through the contralateral visual field, the response is completely abolished (lower trace, Figure 3b [34]). The long-range inhibition persists when the second target is well beyond the region of binocular overlap, in favor of this inhibition being caused by heterolateral interactions [34].

CSTMD1 responses to target motion build up very slowly [39]. It takes several hundred milliseconds for the response to a target that starts its trajectory in the middle of the receptive field (gray trace, Figure 3c), to reach the response level of a space-aligned control where the target traversed a larger part of the receptive field (black, Figure 3c). The slow response onset is independent of where in the classic receptive field the target starts its trajectory [39]. Several hundred milliseconds is an eternity in neuroethological terms, a time frame during which any sensible prey (the most likely target for a predatory dragonfly) would be long gone. It is furthermore not typical of STMDs for example small field-STMDs have very short response delays [38].

What is then the role of such slow response facilitation? An attractive hypothesis is that CSTMD1 serves a role in regulating attention (for recent review on attentional modulation, see [40]). The second output of CSTMD1 arbors across the entire contralateral lobula [35], where it may modify the response, sensitivity, or gain of other STMDs (Figure 3a). Another intriguing possibility is that CSTMD1 inhibits the response of neurons sensitive to optic flow, even though these neurons typically reside in the lobula plate [27]. The rapid optomotor response [41] would otherwise counteract voluntary turns during target pursuit [42], so active suppression has to take place at some stage (see also discussion of this in Review by Olberg in this issue [4]).

Conclusion

While the spatiotemporal processing of target motion in STMDs provides an extremely tantalizing model, it is important not to forget the role of figure detection in the lobula plate. Lobula plate FD cells, which likely underlie forward fixation [24•], generate figure selectivity via GABA-ergic inhibition from the output of optic flow sensitive neurons [43]. Even though FD cells have not been directly shown in *Drosophila*, GABA plays a role in generating the type of figure selectivity involved in bar fixation [44]. Furthermore, even some neurons traditionally seen as optic flow filters respond to the spatiotemporal dynamics within natural scenes, with such called pattern noise [45,46], and to the relative motion of a large bar against self-generated optic flow [47]. Even if the elementary STMD model [20••] stands in striking contrast to other target models, including the FD model for figure selectivity [15], it is thus likely that both lobula and lobula plate circuits play an important role in discriminating the relative motion of figures, features and targets in complex textured surrounds. In this context it is interesting to note that the lamina and lobula plate are probably the ancient ancestral visual neuropils, while the medulla and lobula are later additions [48], suggesting that STMDs evolved after the lobula plate circuits. Future dissociation of visual circuits in *Drosophila* and larger flies will help elucidate the relative role different inputs and higher-order interneurons play in target visualization.

Acknowledgements

This paper is dedicated *in memoriam* of Prof. Em. Kurt Nordström, Uppsala University. The work was funded by VR (2008-2933).

References and recommended reading

Papers of particular interest, published within the period of review, have been highlighted as:

- • of special interest
- •• of outstanding interest

1. Gaffney MF, Hodos W: **The visual acuity and refractive state of the American kestrel (*Falco sparverius*)**. *Vision Res* 2003, **43**:2053-2059.

2. Collett TS, Land MF: **Visual control of flight behaviour in the hoverfly, *Syritta pipiens* L**. *J Comp Physiol A* 1975, **99**:1-66.

3. Olberg RM, Seaman RC, Coats MI, Henry AF: **Eye movements and target fixation during dragonfly prey-interception flights**. *J Comp Physiol A* 2007, **193**:685-693.

4. Olberg RM: **Visual control of prey-capture flight in dragonflies**. *Curr Opin Neurobiol* 2012 doi: 10.1016/j.conb.2011.11.015.

5. Land MF: **Visual acuity in insects**. *Annu Rev Entomol* 1997, **42**:147-177.

6. Wehrhahn C, Poggio T, Bülthof H: **Tracking and chasing in houseflies (Musca)**. *Biol Cybern* 1982, **45**:123-130.

7. Geurten BR, Kern R, Braun E, Egelhaaf M: **A syntax of hoverfly flight prototypes**. *J Exp Biol* 2010, **213**:2461-2475.

8. Collett T: **Visual neurones for tracking moving targets**. *Nature* 1971, **232**:127-130.

9. O'Carroll D: **Feature-detecting neurons in dragonflies**. *Nature* 1993, **362**:541-543.

10. Olberg RM: **Object- and self-movement detectors in the ventral nerve cord of the dragonfly**. *J Comp Physiol A* 1981, **141**:327-334.

11. Hubel DH, Wiesel TN: **Receptive fields, binocular interaction and functional architecture in the cat's visual cortex**. *J Physiol* 1962, **160**:106-154.

12. •• Nordström K, Barnett PD, O'Carroll DC: **Insect detection of small targets moving in visual clutter**. *PLoS Biol* 2006, **4**:378-386.
Whereas earlier models for figure selectivity rely on a velocity difference signal from the background, this paper showed that hoverfly STMDs respond robustly to target motion even against a moving background, and whether or not there is a velocity difference between the two.

13. Nordström K, O'Carroll DC: **Feature detection and the hypercomplex property in insects**. *Trends Neurosci* 2009, **32**:383-391.

14. Reichardt W, Poggio T, Hausen K: **Figure-ground discrimination by relative movement in the visual system of the fly. Part II: towards the neural circuitry**. *Biol Cybern* 1983, **46**:1-30.

15. Egelhaaf M: **On the neuronal basis of figure-ground discrimination by relative motion in the visual system of the fly. III. Possible input circuitries and behavioral significance of the FD-cells**. *Biol Cybern* 1985, **52**:267-280.

16. Kimmerle B, Egelhaaf M: **Performance of fly visual interneurons during object fixation**. *J Neurosci* 2000, **20**:6256-6266.

17. Egelhaaf M: **On the neuronal basis of figure-ground discrimination by relative motion in the visual system of the fly. II. Figure-detection cells, a new class of visual interneurones**. *Biol Cybern* 1985, **52**:195-209.

18. • Wiederman SD, O'Carroll DC: **Discrimination of features in natural scenes by a dragonfly neuron**. *J Neurosci* 2011, **31**:7141-7144.
In this ingenious paper the authors show that not only are false positive targets surprisingly rare in natural scenes, but also that the ESTMD model correctly predicts which features in a natural scene will evoke a physiological response from STMDs.

19. Collett T, King AJ: **Vision during flight**. In *The Compound Eye and Vision of Insects.* Edited by Horridge GA. Clarendon Press; 1975: 467-466.

20. •• Wiederman SD, Shoemaker PA, O'Carroll DC: **A model for the detection of moving targets in visual clutter inspired by insect physiology**. *PLoS One* 2008, **3**:e2784.
This important paper provides a biomimetic model that can account for the sharp size tuning of STMDs, and their ability to respond robustly to targets in clutter. It is based on a series of clever spatial and temporal filters generating sharp tuning to the unique spatial profile of a small moving target.

21. Strausfeld NJ: *An Atlas of an Insect Brain*. Berlin: Springer-Verlag; 1976.

22. Fischbach K, Dittrich A: **The optic lobe of *Drosophila melanogaster*. Part I: a Golgi analysis of wild-type structure**. *Cell Tissue Res* 1989, **258**:441-475.

23. Zhu Y, Nern A, Zipursky SL, Frye MA: **Peripheral visual circuits functionally segregate motion and phototaxis behaviors in the fly**. *Curr Biol* 2009, **19**:613-619.

24. • Maimon G, Straw AD, Dickinson MH: **A simple vision-based algorithm for decision making in flying *Drosophila***. *Curr Biol* 2008, **18**:464-470.
This brilliant study uses behavior to show that optic flow, and the motion of small and large objects, are processed in parallel computational streams. The study thus provides a behavioral account for the notion that motion vision is processed in a series of partially non-overlapping input pathways.

25. Hassenstein B, Reichardt W: **Systemtheoretische Analyse der Zeit, Reihenfolgen und Vorzeichenauswertung Bei der Bewegungsperzeption des Rüsselkafers *Chlorophanus***. *Z Naturforsch* 1956, **11**:513-524.

26. Borst A, Euler T: **Seeing things in motion: models, circuits, and mechanisms**. *Neuron* 2011, **71**:974-994.

27. Borst A, Haag J, Reiff DF: **Fly motion vision**. *Annu Rev Neurosci* 2010, **33**:49-70.

28. Clark DA, Bursztyn L, Horowitz MA, Schnitzer MJ, Clandinin TR: **Defining the computational structure of the motion detector in *Drosophila***. *Neuron* 2011, **70**:1165-1177.

29. Eichner H, Joesch M, Schnell B, Reiff DF, Borst A: **Internal structure of the fly elementary motion detector**. *Neuron* 2011, **70**:1155-1164.

30. Gabbiani F, Jones PW: **A genetic push to understand motion detection**. *Neuron* 2011, **70**:1023-1025.

31. Reiff DF, Plett J, Mank M, Griesbeck O, Borst A: **Visualizing retinotopic half-wave rectified input to the motion detection circuitry of *Drosophila***. *Nat Neurosci* 2010, **13**:973-978.

32. • Joesch M, Schnell B, Raghu SV, Reiff DF, Borst A: **ON and OFF pathways in *Drosophila* motion vision**. *Nature* 2010, **468**:300-304.
Using genetic tools in combination with technically challenging electrophysiology the authors show that the input to the EMD is split into separate on and off channels, via the LMCs L1 and L2, respectively.

33. Franceschini N, Riehle A, Le Nestour A: **Directionally selective motion detection by insect neurons**. In *Facets of Vision.* Edited by Stavenga DG, Hardie RC. Springer-Verlag; 1989:360-390.

34. Bolzon DM, Nordström K, O'Carroll DC: **Local and large-range inhibition in feature detection**. *J Neurosci* 2009, **29**:14143-14150.

35. Geurten BRH, Nordström K, Sprayberry JDH, Bolzon DM, O'Carroll DC: **Neural mechanisms underlying target detection in a dragonfly centrifugal neuron**. *J Exp Biol* 2007, **210**:3277-3284.

36. Nordström K, O'Carroll DC: **Small object detection neurons in female hoverflies**. *Proc R Soc Lond B* 2006, **273**:1211-1216.

37. Olberg RM: **Identified target-selective visual interneurons descending from the dragonfly brain**. *J Comp Physiol A* 1986, **159**:827-840.

38. Barnett PD, Nordström K, O'Carroll DC: **Retinotopic organization of small-field-target-detecting neurons in the insect visual system**. *Curr Biol* 2007, **17**:569-578.

39. Nordström K, Bolzon DM, O'Carroll DC: **Spatial facilitation by a high-performance dragonfly target-detecting neuron**. *Biol Lett* 2011, **7**:588-592.

40. Maimon G: **Modulation of visual physiology by behavioral state in monkeys, mice, and flies**. *Curr Opin Neurobiol* 2011.

41. Theobald JC, Ringach DL, Frye MA: **Dynamics of optomotor responses in *Drosophila* to perturbations in optic flow**. *J Exp Biol* 2010, **213**:1366-1375.

42. Trischler C, Kern R, Egelhaaf M: **Chasing behavior and optomotor following in free-flying male blowflies: flight performance and interactions of the underlying control systems**. *Front Behav Neurosci* 2010, **4**:20.

43. Warzecha A-K, Egelhaaf M, Borst A: **Neural circuit tuning fly visual interneurons to motion of small objects. I. Dissection of the circuit by pharmacological and photoinactivation techniques**. *J Neurophysiol* 1993, **69**:329-339.

44. Fei H, Chow DM, Chen A, Romero-Calderon R, Ong WS, Ackerson LC, Maidment NT, Simpson JH, Frye MA, Krantz DE: **Mutation of the *Drosophila* vesicular GABA transporter disrupts visual figure detection**. *J Exp Biol* 2010, **213**:1717-1730.

45. Meyer HG, Lindemann JP, Egelhaaf M: **Pattern-dependent response modulations in motion-sensitive visual interneurons — a model study**. *PLoS One* 2011, **6**:e21488.

46. O'Carroll DC, Barnett PD, Nordström K: **Local and global responses of insect motion detectors to the spatial structure of natural scenes**. *J Vis* 2011, **11**: paper 20.

47. Liang P, Kern R, Egelhaaf M: **Motion adaptation enhances object-induced neural activity in three-dimensional virtual environment**. *J Neurosci* 2008, **28**:11328-11332.

48. Strausfeld NJ: **Brain organization and the origin of insects: an assessment**. *Proc Biol Sci* 2009, **276**:1929-1937.